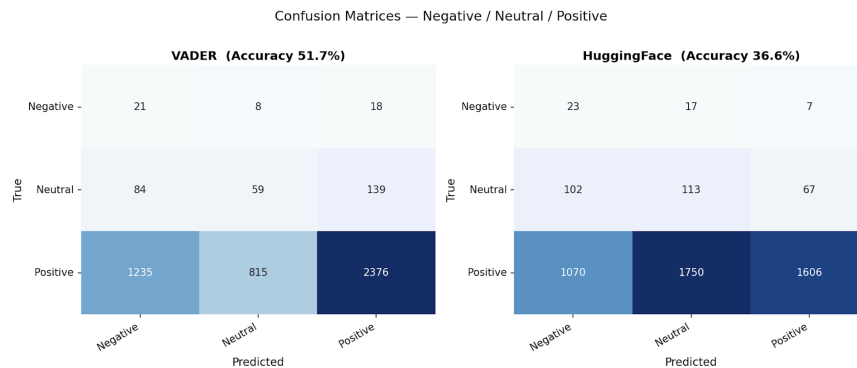


Do Movie Reviews Actually Say What They Mean?

Comparing Sentiment Analysis Models on Real-World Film Reviews



Every day, millions of people log onto Letterboxd to share their thoughts on the movies they watch. Some write a single sentence. Some write paragraphs. Some use emojis, sarcasm, or inside jokes. They also leave a star rating, a simple number from half a star to five. On the surface, the rating and the review should tell the same story. But do they?

Sentiment analysis is widely used in data science, from product review monitoring to social media trend detection. The idea is simple: given a piece of text, determine whether the sentiment behind it is positive, negative, or neutral. But in practice, the choice of model matters enormously. A rule-based approach like VADER, which works by matching words against a crafted lexicon of sentiment scores, behaves very differently from a transformer model like HuggingFace, which was trained on hundreds of millions of social media posts to learn. Both are widely used. But which one actually works better on the kind of informal, emotionally complex text that real people write when they talk about movies?

Using a dataset of over 4,700 Letterboxd reviews from the platform's Top 250 films, you will run both VADER and a pretrained HuggingFace model and compare each model's sentiment predictions against the star ratings left by the same reviewers. The star rating is your ground truth and your analysis will tell you how often each model agrees with that truth, where each model fails, and what those failures reveal about the limits of sentiment analysis on real-world text.

By running both models and evaluating their outputs against the star-rating ground truth, you will be able to determine for yourself which approach better captures sentiment in informal, review-style writing, and more importantly, why each model succeeds or fails. This case study will give you hands-on experience with pretrained NLP models, model evaluation, and critical thinking about tool selection that can be applicable in typical data science work.

GitHub Repo: <https://github.com/jvanbenschoten/CS3-DS4002-Sp26.git>

Letterboxd Top 250 Films Reviews — provided in the DATA/ folder of the repo.